

# LD10 Assignment Proof Checklist

Evidence that required practical tasks were completed

## 1. Checklist proof

| Level        | Required task                                                                                                  | Notebook evidence                                                                 | Status  |
|--------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------|
| Beginner     | Install/import XGBoost, load Bank Marketing, encode categorical columns with LabelEncoder, train XGBClassifier | Completed in classification section                                               | Covered |
| Beginner     | Compare Accuracy and F1-score of XGBClassifier vs RandomForestClassifier                                       | XGB F1 0.8499 vs RF F1 0.8441                                                     | Covered |
| Beginner     | Plot feature importances and identify strongest deposit driver                                                 | Top feature: duration (0.1820)                                                    | Covered |
| Intermediate | Define XGBoost grid and run GridSearchCV                                                                       | Best params: max_depth=3, learning_rate=0.1, n_estimators=100                     | Covered |
| Intermediate | Train CatBoostRegressor without manual encoding and compare RMSE                                               | CatBoost RMSE \$26,664.89                                                         | Covered |
| Intermediate | Enable early stopping and show optimal iteration                                                               | CatBoost best iteration: 442                                                      | Covered |
| Expert       | Implement Optuna study for LightGBM                                                                            | 10 trials executed in FAST_MODE; full 50-trial path available                     | Covered |
| Expert       | Visualise Optuna results                                                                                       | Optimization history, parameter importances and parallel coordinate plots created | Covered |
| Expert       | Create averaging ensemble of LightGBM, XGBoost and CatBoost                                                    | Ensemble RMSE \$25,495.43                                                         | Covered |

### Checklist conclusion

All required assignment tasks are represented in the final master notebook. The notebook also includes additional theory and professional interpretation beyond the checklist.

## 2. Extra work beyond the checklist

- Full source audit and coverage matrix
- Bagging vs Boosting comparison
- Random Forest, OOB and bias-variance recap
- AdaBoost and Gradient Boosting mini-demos
- Forward stagewise additive modelling and pseudo-residual theory
- Professional BI-style charts and business conclusions
- Variable and method flow documentation